

Libyan Academy

ESP pump failure Risk Analysis

Risk Management



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Contents

Abstract:	2
Introduction:	3
Literature reviews:	4
Case study Overview:	5
Case study Description:	5
Risk Management Methodologies:	6
Risk Identification:	7
Overview of identifying Risk in ESP Pump Failure:	7
Pump parties that can damage and cause ESP failure:	9
Qualitative risk analysis:	11
Quantitative risk analysis:	12
Reservoir or fluids risks:	12
Reservoir or fluids risks distribution:	13
Reservoir or Fluids Plan Risks Responses and Resulting:	14
Wear-and-Tear Risks:	15
Wear-and-Tear risks distribution:	17
Wear-and-Tear Plan Risks Responses and Resulting:	17
System design/Selection Risks:	18
System design/Selection risks distribution:	19
System design/Selection Plan Risks Responses and Resulting:	20
Installation Risks:	21
Installation risks distribution:	23
Installation Plan Risks Responses and Resulting:	23
Manufacturing Risks:	24
Manufacturing risks distribution:	25
Manufacturing Plan Risks Responses and Resulting:	26
Conclusion:	28
References:	29

Abstract:

In oil industry, ESP (Electric Submersible Pump) failure risks play a significant part in decision-making and may affect the performance of the production and additional operation costs. Moreover, ESP always face uncertainty that leads to cost and schedule risks. Most of those risks can be mitigated with better up-front planning and analysis.

In this study, risks associated to more than 282 ESPs operating at Akakus Oil Company in El- Sharara field, producing fluids from three concession, over time the number of production wells has increased, and additional ESPs have brought on line, which make the number of failures increasing with time. The qualitative risk analysis process carried results by risk 3x3 matrix that helped us to rate the significance of our identified risks based on the probability of the risk materializing and the impact if it does. The results showed a reduction on ESP risk failures and risk responses reduce the action of likelihood of occurrence or the impact of risks.

Introduction:

Electric Submersible Pump (ESP) failures have important economic implications for O&G operators, including nonproductive time, production deferral, production losses, and high cost of ESP replacements. Since the birth of ESP in 1926 the technology has not evolved significantly and the technology of today has many similarities with the original ESP system designed in 1926. Today approximately 10% of the world oil production is produced with electrical submerged pumping. ESP are also known for lifting much greater liquid rates than most of the other types of artificial lift, both on- and offshore. Considering the large number of ESPs deployed in an oilfield, the bulk of ESP failure prediction research has focused on computing and estimating causes of failure. Statistical analysis of ESP failures across an oilfield has shown that the failure rates vary considerably.

Submersible pumps (ESPs) consist of multiple centrifugal pump stages mounted in series within a housing mated to a submersible electric motor. These pumps connected to surface controls and electric power by armor-protected cables (Bernt H, 2010).

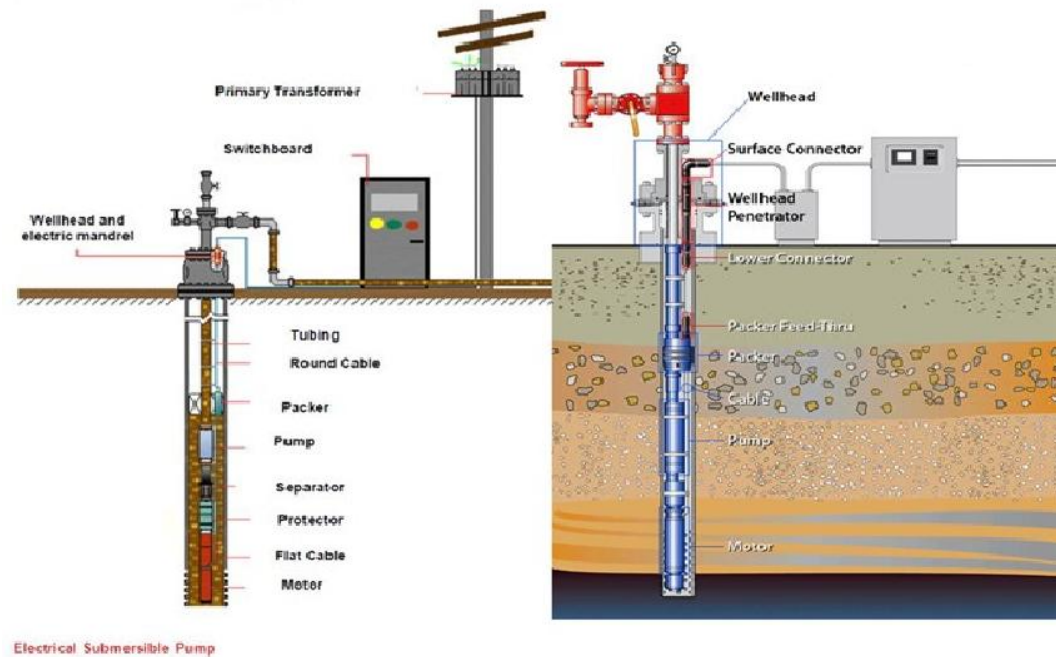


Figure (1)

There are more than 282 ESPs operating at Akakus Oil Company in El- Sharara field, producing fluids from three concession, over time the number of production wells has increased, and additional ESPs have been brought on line, which make the number of failures increasing with time. The objectives of this study is:

- To Identify and clarify the possible causes of ESP failure and the best way to avoid the failures.
- To provide better ESP performance and increase the run life as long as possible by doing Risk Analysis.
- To reduce the cost of ESP repairing and ESP inventory can be minimized which saves both money and space (Mohamed D, 2018).

Literature reviews:

Many previous studies have been conduct within the field of integrated cost and schedule risk analysis but each presents a different approach to this concept. In 2010 Hollund studied the Artificial Lift – Electrical Submerged Pump, best practice and future demands within subsea applications on Peregrino field USA using a

survey of ESP failures conducted in the database ESP-RIFTS. In addition is a LCC analysis performed, that identified work-over costs as the most critical cost drivers for Peregrino field.

Alhanati F.J.S et al. (2001) states an approach to establish consistent practices for collecting, tracking and sharing ESP run life and failure information based in two elements a general and common data set and a standard nomenclature for coding ESP failure information. The general data set contains basic information on operating conditions, ESP equipment, and the observed failures.

Om Prasad P et al. (2014) states how a time-series analysis approach applied to failure detection and failure prediction from the streams of sensor data. The method involves identifying “shapelets” – short instances that are particularly distinct – in the streams of sensor data.

Hisham A et al, (2003) states performance of Electrical Submersible Pumps (ESPs) in the Wafra field within the Divided Zone Kuwait-Saudi Arabia and covers the period of 4 years started in 1998-December 2001. Included are the historical failures of ESPs, Analysis of failures and some solutions that have helped to reduce the number of failures and that in-turn increased ESP run lives/reduced deferred oil production/saved rig cost/time.

Case study Overview:

Akakus oil Operations Runs a number of key fields all these fields deployed on a large area of desert land connected to the main pipeline for crude oil tankers as well as network lines to transfer the electrical power needed for production operations and depends on ESP Pumps.

In addition, System design selection for ESP pump, equipment specification, manufacturing, installation, and day-to-day operation can be identify and corrected, contributing to increased run lives, lower operating costs and increased profits. Because the company has the full financial and experience potential to mitigate this risk, could create a team to study and monitor this risk.

Case study Description:

Akakus Oil Operation Company works with Schlumberger Service Company to provide and install ESP Pumps. **In this study**, the cost risks associated to oil field called A -Field, it consists of 18 wells according to experts and high knowledge from Schlumberger Company and Akakus Oil Company Workover Office. Because of A-Field has a good oil production, and critical in its urgent operations. It has a PVT parameters as it following (Muhand S,2018):

Initial Pressure (psia)	1958
Reservoir Temperature (°F)	204f = 95.55 °C
Bubble Point (psig)	219
GOR %	64
Oil Rate bbl/d	86570
Pressure (psi)	1182.71

Risk Management Methodologies:

Project Management Institute (PMI) defined risk as an uncertain event or condition that, if it occurs, has a positive or negative effect on one or more project objectives such as scope, schedule, cost, and quality (PMBOK Guide, 2013). A risk may have one or more causes and, if it occurs, it may have one or more impacts. A cause may be a given or potential requirement, assumption, constraint, or condition that creates the possibility of negative or positive outcomes. For example, causes could include the requirement of an environmental permit to do work, or having limited personnel assigned to design the project.

The study used different ways to identify risks, such as documentation review, daily reports, ESP pull reports, information gathering (Brainstorming with drilling and production engineers, interviewing with Schlumberger supervisor, petroleum active supervisor, and Production supervisor at Akakus Company). Figure (2) shows the methodologies that used in this study.

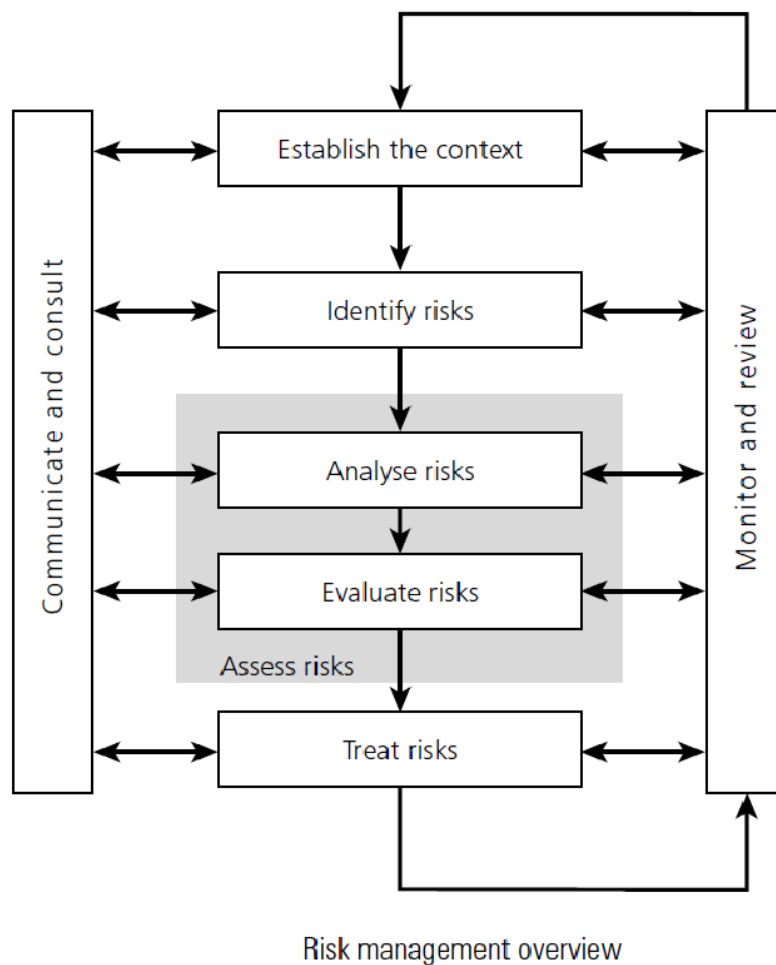
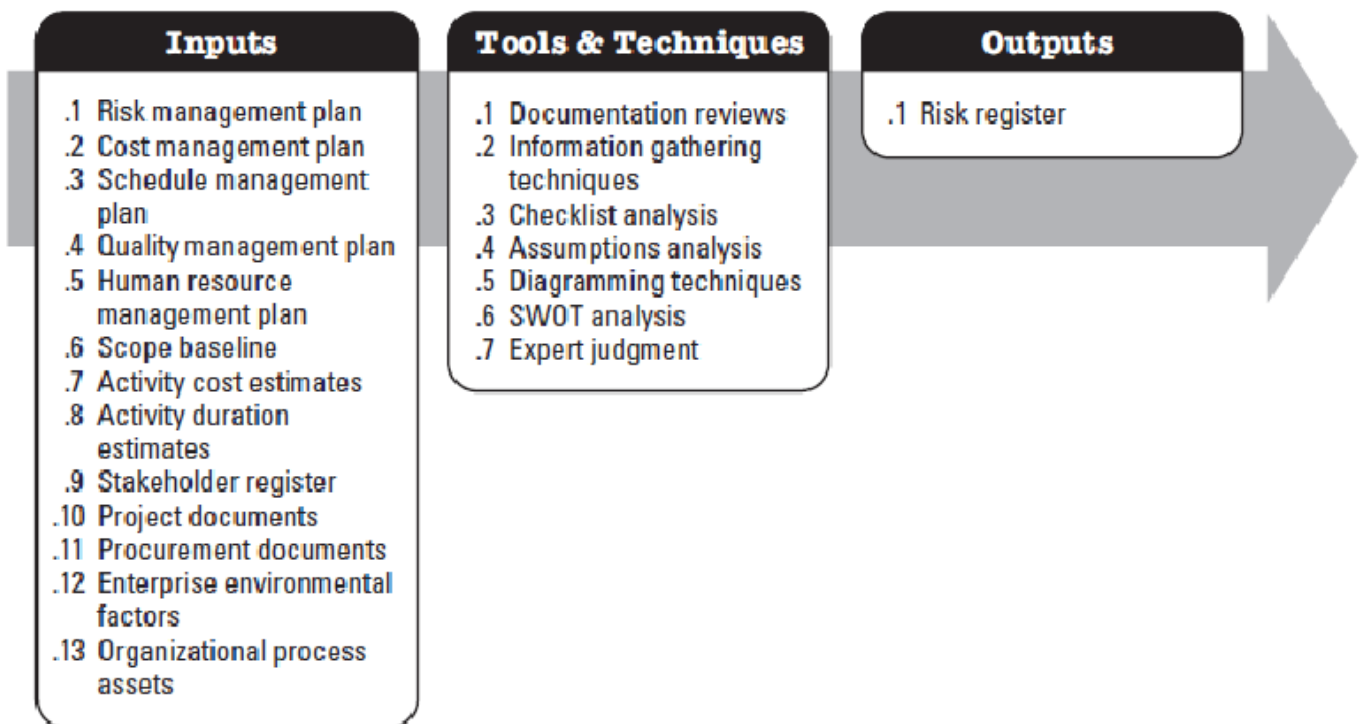


Figure (2) (Joint Technical Committee, 1999)

Risk Identification:

Identify Risks is the process of determining which risks may affect the project and documenting their characteristics. The key benefit of this process is the documentation of existing risks and the knowledge and ability it provides to the project team to anticipate events. Participants in risk identification activities may include the following: project manager, project team members, risk management team (if assigned), customers, subject matter experts from outside the project team, end users, other project managers, stakeholders, and risk management experts. (PMBOK Guide, 2013).



Identify Risks: Inputs, Tools & Techniques, and Outputs

Figure (3)

Overview of identifying Risk in ESP Pump Failure:

Risks related to Completion Failure:

1. Failure of perforations or liner.
2. Failure in the sand control system.

3. Wellbore completion failure.
Failure of the wellbore completion for example:
Casing, tubing, packer, safety valve or liner failures.

Risks related to Installation:

1. System assembly such as Mounting of the equipment.
2. Reran damaged equipment.
3. Well cleanout.
4. ESP field service such as Testing Well optimization.

Risks related to Manufacturing:

1. Wrong material selection.
2. Improper fabrication or assembly of Components.
3. Improper equipment testing or quality control.

Risks related to Operation:

1. Inadequate condition monitoring.
2. Poor operating procedure for example if it is performed unnecessary many start/stop.
3. Well treatment like Injection of inhibitors.

Risks related to Wear-and-Tear:

1. Pump worn, Broken or fractured.
2. Motor contaminated, corroded.
3. Cable low resistance.
4. Seal contaminated, corroded.
5. Pump Intake, Broken or fractured.

Risks related to Reservoir or Fluids:

1. Presence of asphaltene in the well fluid.
2. Bottom-hole temperature above approximately 80°C.
3. Free gas in well fluid.
4. Sand content in well fluid.
5. Scale formation.
6. High water cut in the well fluid.
7. Low or no inflow of fluids in the well.

Risks related to System Design / Selection:

1. Wrong selection of ESP equipment for the given application such as.
 - Inadequate pump flow or head capacity.
 - Motor power capacity
 - Use of low-grade materials
2. Improper system design / selection, including use of improper data or errors in calculations.

Pump parties that can damage and cause ESP failure:

Electrical Cable:

The possible causes of cable failures are:

- Mechanical damage (crushing, cutting, etc.) during running or pulling operations.
- Corrosion.
- Insulation deterioration due to high temperature or wellbore gases.
- Currents above the design limit can rapidly increase cable temperature leading to insulation breakdown.

PUMP:

Typical root causes to pump failures listed in the following:

- Up-thrust wear occurs when the pump operated at flow rates greater than the maximum recommended flow rate. In floating-type pumps the impellers forced up against the diffusers, the up thrust washers may be overloaded and it can result in a destroyed pump stage. Solids production accelerates the process.
- Down-thrust wear typically occurs at flow rates lower than the minimum recommended rate also accelerated by solids production. Down-thrust washers destroyed after the pump stage destroyed.
- Scale formation can plug or even lock pump stages.

Motor:

1. **Overloading** of the motor leads to heating of the motor's wiring; can lead to damage or burnout. Overload is usually caused by:
 - Undersized motor.
 - High specific gravity of the well fluid can cause the actual total dynamic head (TDH) to be greater than assumed.
 - Pump failures that result in increased power requirement.
 - Irregular (high, low, or unbalanced) voltage at the motor terminal.
2. **Leaking** protectors allow well fluids to enter the motor causing a gradual contamination of the motor oil, which can lead to short circuit or motor burnout.
3. **Insufficient cooling.** Cooling is insufficient when; flow velocity past the motor is lower than required for an efficient transfer of the generated heat from the motor.

Seal Failures:

Seal section failures may attributed to the following factors:

1. Broken or damaged mechanical seals in the seal section can cause well fluids to leak into the motor. This can be a result of:
 - Vibrations transmitted from a worn pump.
 - Wrong equipment selection or improper installation.
2. The ESP main thrust bearing in the seal may fail when the pump operates in excessive up or down-thrust mode.
3. Labyrinth-type seals may fail in deviated (more than 30 degrees) well sections.

Shaft Failures:

1. Torsional yield failure occurs when the torque limit of the shaft exceeded. Can result in permanent deformation or a break of the shaft.
2. Torsional twist in the shaft absorbs the energy during system start-up. If the shaft twists more than allowed by a connected part then that part can be damage.
3. Bearing wear mostly occurs when solids produced when clearances in radial bearings increase. This will weaken the shaft stability and vibrations will increase.
4. Lost lubrication in radial bearings of the pump can happened because of the metal-to-metal rubbing create increased local temperatures, and bearing materials weld together. This can lead to a damaged or broken shaft, (Bernt H, 2010).

Causes of ESP pump failures that exist in Akakus Oil Operation fields according to ODH statistics and daily reports. It listed as categories and sub-categories in Risk Breakdown Structure (RBS), (Abo Baker E,2018):

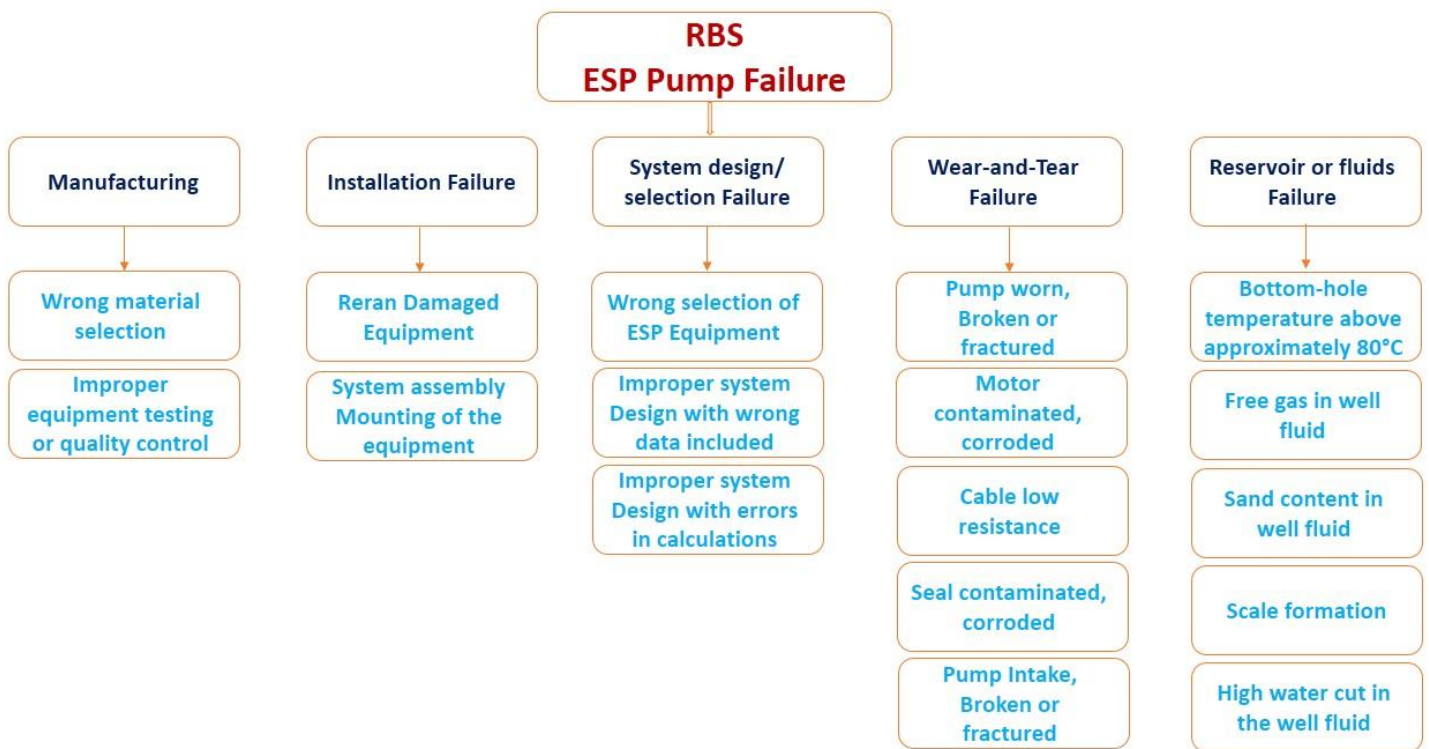


Figure (4)

Qualitative risk analysis:

Qualitative Risk Analysis is the process of prioritizing risks for further analysis or action by assessing and combining their probability of occurrence and impact. The key benefit of this process is that it enables project managers to reduce the level of uncertainty and to focus on high-priority risks (PMBOK Guide, 2013).

Probability Rating	>60% Probable	3 High/ Critical	3	6	9
	20-60% Could happen	2 Medium/ Serious	2	4	6
	<20% Improbable	1 Low/ Marginal	1	2	3
			1 Low/Improbable	2 Medium/ Could happen	3 High/ Probable
			Cost Impact	<10% cost increase	10-20% cost increase
					> 20% cost increase
			Impact Rating		
Risk Action					
	Key/ Critical Risks -Immediate program-level attention needed				Risk Rating
	Intermediate Risks - Project-level planning required				1-2 Low
	Minor Risks - Manage by normal procedures				3-4 Medium
					5-9 High

Probability and Impact Matrix

Table (1)

The risk 3x3 matrix helps us to rate the significance of our identified risks based on the probability of the risk materializing and the impact if it does. There are various sizes and types of risk matrix, depending on the rating scale chosen. The simplest

form uses a 3×3 grid, but you could use a 4×4, a 5×5 or a 6×6 grid, depending on how complicated or simple you want to make things. In reality, it does not really matter. Because when we look at probability and impact, we are really trying to determine the most significant risks, in other words, the ones that are most likely to happen and the ones with the most serious consequences.

Quantitative risk analysis:

Quantitative Risk Analysis is the process of numerically analyzing the effect of identified risks on overall project objectives. The key benefit of this process is that it produces quantitative risk information to support decision making in order to reduce project uncertainty (PMBOK Guide, 2013).

This is a numerical analysis of the probability (P) and impact (I) of the risk on the project. Determine overall project risk (risk exposure), identify risks requiring the most attention, determine cost and schedule reserves, and create realistic and achievable cost, schedule, or scope targets (Tawfik R et al, 2015).

$$\text{Risk Score Value (Rating)} = P \times I$$

Reservoir or fluids risks:

Reservoir or fluids risks are the risks affecting the ESPs objectives and goals.

They include risks arising:

- ✓ Bottom hole temperature above 80c.
- ✓ Free gas in the well fluid.
- ✓ Sand content in well fluid.
- ✓ Scale formation.
- ✓ High water cut in the well fluid.

By using the 3 x 3 Risk Matrix shown in Table 1 we will assess and evaluate risks based on the probability of occurrence and the impact on the ESPs, we will also assign responsibilities to manage these risks by making residual risks after mitigation actions.

Risk Identification			Risk Analysis			Response Planning			Residual Risk after mitigation actions		
Category	No.of risk	Risk	P	I	Risk Score	Risk Strategy	Responsibility	Risk Mitigation & Actions to resolve	P	I	Risk Score
Reservoir or Fluids	1	Bottom hole temperature above 80c	2	2	4	Mitigate	ESP design section head	Select a Motor that can handle the well temperature.	1	2	2
	2	Free gas in the well fluid	2	3	6	Mitigate	ESP design section head	Select a cable with proper insulation.	1	2	2
	3	Sand content in well fluid	1	2	2	Mitigate	ESP design section head	Selecting a proper pump type and materials.	1	3	3
	4	Scale formation	2	2	4	Mitigate	Workover section	Running bit scrubber before running ESP Pump.	1	1	1
	5	High water cut in the well fluid	2	2	4	Mitigate	ESP design section head	Select motor and pumps with the proper specifications that capable of deliver the desired flow rate.	1	1	1

Table (2)

Reservoir or fluids risks distribution:

Figure (5) below shows the distribution of the Reservoir or fluids risks based on the ratings generated from Table (Reservoir or fluids Risk Assessment).

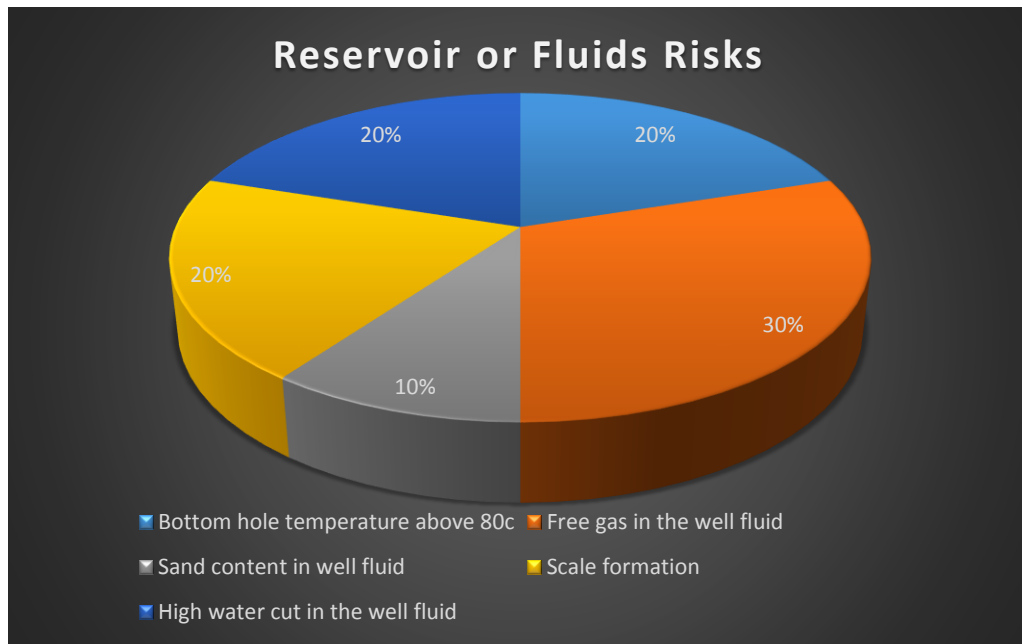


Figure (5)

Reservoir or Fluids Plan Risks Responses and Resulting:

Reservoir or fluids risks responses defined as the set of actions that seek to reduce the likelihood of occurrence or the impact of risks, we will examine the different categories of actions and controls that can employed.

Bottom hole temperature above 80c: High well temperatures can have detrimental effect on the motor, which can overheat and burn out. Here it is important to select a pump that can handle the well temperature.

Free gas in the well fluid: Formation gas can lead to gas lock in the pump. By utilize proper gas handlers or separators one will be able to a certain degree withstands this threat. Free gas can also migrate into the power cable; to mitigate this one can select a cable with proper insulation.

Sand content in well fluid: Sand or abrasive production problems can be avoid or reduce by selecting a proper pump type and materials.

Scale formation: Scale deposits on motor housings will obstruct the motor in getting enough cooling effect from the well fluid, which can result in motor burn out. Special coatings applied to the motor housing are the usual solution. Scale formation in pump stages reduces the liquid rate and can result in a plugged pump.

High water cut in the well fluid: To reduce this problem Select motor and pumps with the proper specifications that capable of deliver the desired flow rate.

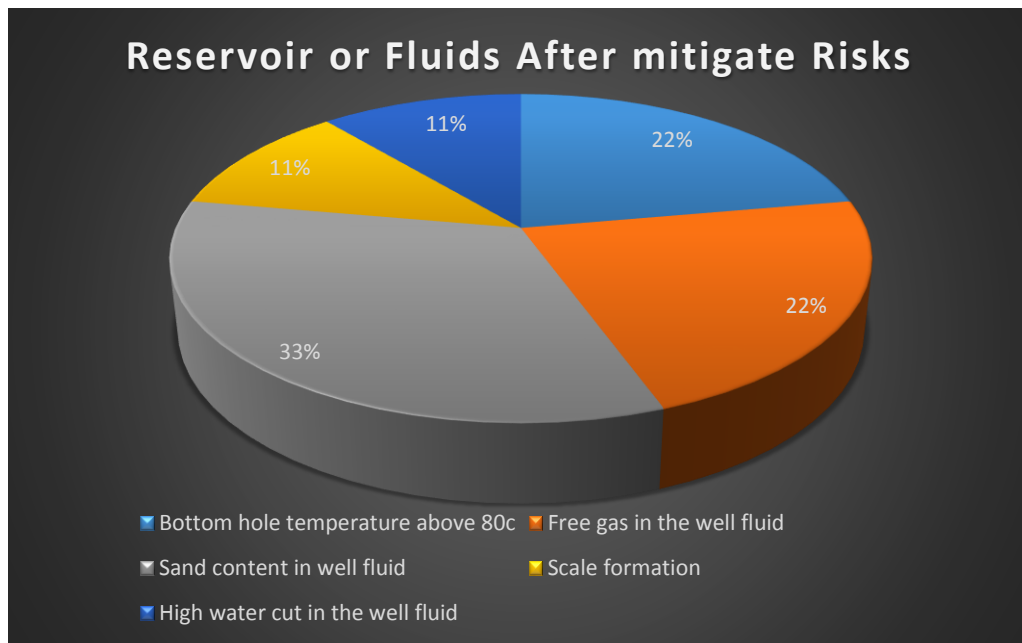


Figure (6)

Wear-and-Tear Risks:

Wear-and-Tear risks are the risks affecting the ESPs objectives and goals.

They include risks arising:

- ✓ Motor contaminated, corroded.
- ✓ Cable low resistance.
- ✓ Seal contaminated, corroded.
- ✓ Pump worn, broken or fractured.
- ✓ Pump Intake, broken or fractured.

By using the 3 x 3 Risk Matrix shown in Table 1 we will assess and evaluate risks based on the probability of occurrence and the impact on the ESPs, we will also

assign responsibilities to manage these risks by making residual risks after mitigation actions.

Risk Identification			Risk Analysis			Response Planning			Residual Risk after mitigation actions		
Category	No.of risk	Risk	P	I	Risk Score	Risk Strategy	Responsibility	Risk Mitigation & Actions to resolve	P	I	Risk Score
Wear-and-Tear	1	Motor contaminated, corroded	2	2	4	Mitigate	ESP design section head	Select proper material properties to handle the harsh environments.	1	1	1
	2	Cable low resistance	2	3	6	Mitigate	ESP vendor	Cables must be durable in a wide range of conditions, and ESP components have been tested for proper operation before it is installed in the well.	1	3	3
	3	seal contaminated, corroded	1	2	2	Mitigate	ESP design section head	Select Seal with proper materials that handle corrosion.	1	1	1
	4	Pump worn, broken or fractured	1	2	2	Mitigate	Workover section & ESP design section head	Sand control to handle sand production and select pump with the proper specifications that capable of deliver the desired flow rate.	1	1	1
	5	Pump Intake, broken or fractured	1	2	2	Mitigate	Workover section	Sand control to handle sand production.	1	2	2

Table (3)

Wear-and-Tear risks distribution:

Figure (6) below shows the distribution of the Wear-and-Tear risks based on the ratings generated from Table (Wear-and-Tear Risk Assessment).

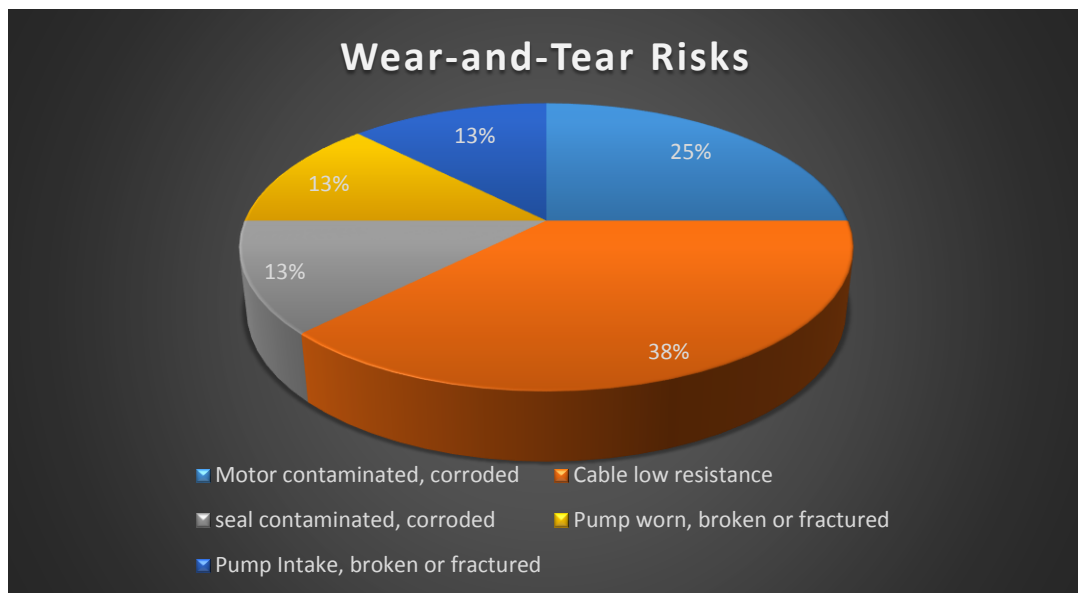


Figure (7)

Wear-and-Tear Plan Risks Responses and Resulting:

Wear-and-Tear risks responses defined as the set of actions that seek to reduce the likelihood of occurrence or the impact of risks, we will examine the different categories of actions and controls that can employed.

Motor contaminated, corroded: Here it is important to select proper material properties to handle the harsh environments.

Cable low resistance: When an ESP subjected to severe sand production, cable has a significant failure rate than the other ESP equipment.to reduce this Cables must be durable in a wide range of conditions, and ESP components have been tested for proper operation before it is installed in the well.

Seal contaminated, corroded: To mitigate this one can select Seal with proper materials that handle corrosion.

Pump worn, broken or fractured: If we compare this result with ESP that have minimum produced solids at all then we see that it is a more “normal” distribution in the failure rate among the different equipment. From this, we can conclude with that solids have a bad influence on pump lifetime. To reduce this Sand control to handle sand production and select pump with the proper specifications that capable of deliver the desired flow rate.

Pump Intake, broken or fractured: Since an ESP system placed at the bottom of the production string; it can be sucked large quantities of sand into the pump intake at startup. Filters or screens must utilized.

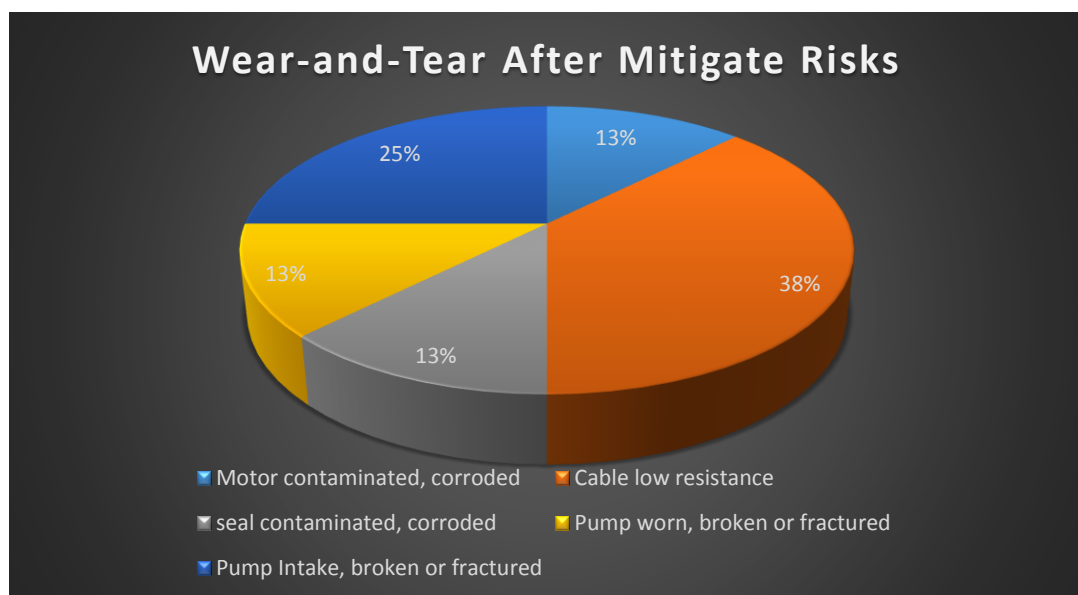


Figure (8)

System design/Selection Risks:

System design/Selection risks are the risks affecting the ESPs objectives and goals.

They include risks arising:

- ✓ Wrong selection of ESP equipment.
- ✓ Improper system design with wrong data included.
- ✓ Improper system design with errors in calculation.

By using the 3 x 3 Risk Matrix shown in Table 1 we will assess and evaluate risks based on the probability of occurrence and the impact on the ESPs, we will also assign responsibilities to manage these risks by making residual risks after mitigation actions.

Risk Identification			Risk Analysis			Response Planning			Residual Risk after mitigation actions		
Category	No.of risk	Risk	P	I	Risk Score	Risk Strategy	Responsibility	Risk Mitigation & Actions to resolve	P	I	Risk Score
System design/Selection	1	Wrong selection of ESP equipment	1	3	3	Mitigate	ESP design section head	Select proper ESP equipment installation design.	1	1	1
	2	Improper system design with wrong data included	2	2	4	Mitigate	ESP design section head & petroleum section	Input reliable data on the well productivity and accurate information on fluid properties like viscosity, GOR, and temperature.	1	1	1
	3	Improper system design with errors in calculation	2	2	4	Mitigate	ESP design section head	Knowledgeable personnel that make calculation on ESP design selection software.	1	2	2

Table (4)

System design/Selection risks distribution:

Figure (7) below shows the distribution of the System design/Selection risks based on the ratings generated from Table (System design/Selection Risk Assessment).

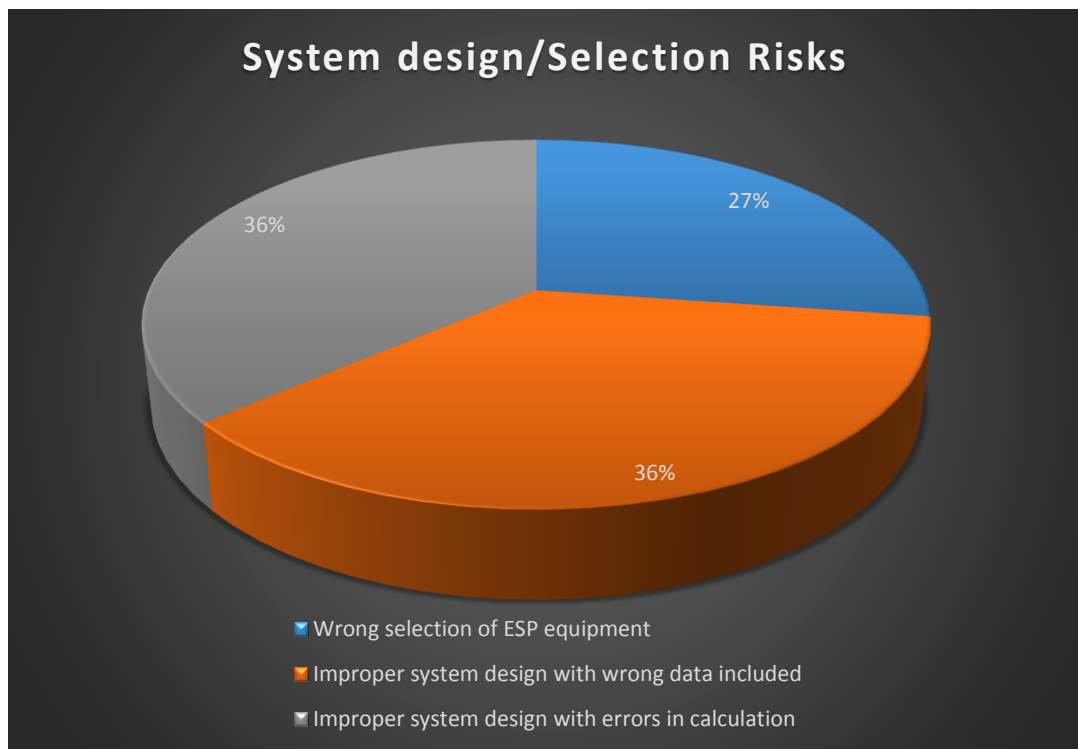


Figure (9)

System design/Selection Plan Risks Responses and Resulting:

System design/Selection risks responses defined as the set of actions that seek to reduce the likelihood of occurrence or the impact of risks, we will examine the different actions and controls that can employed.

Wrong selection of ESP equipment, improper system design with wrong data included and improper system design with errors in calculation all of these critical failures can increase costs and effect on objective and goals of the operations. It is important to conduct a proper installation design to prevent against system design /selection failures. Initial sizing of ESP equipment is the most important factor to succeed in optimizing the run life of ESP. This implies that the unit must be operating within the recommended operation window with regarding to flow range, temperature, GOR etc. If the ESP run outside its operation window, pump wear accelerated leading to premature pump and motor failures. Reliable data on the

well productivity and accurate information on fluid properties like viscosity, GOR, and temperature is important to be able to select proper equipment.

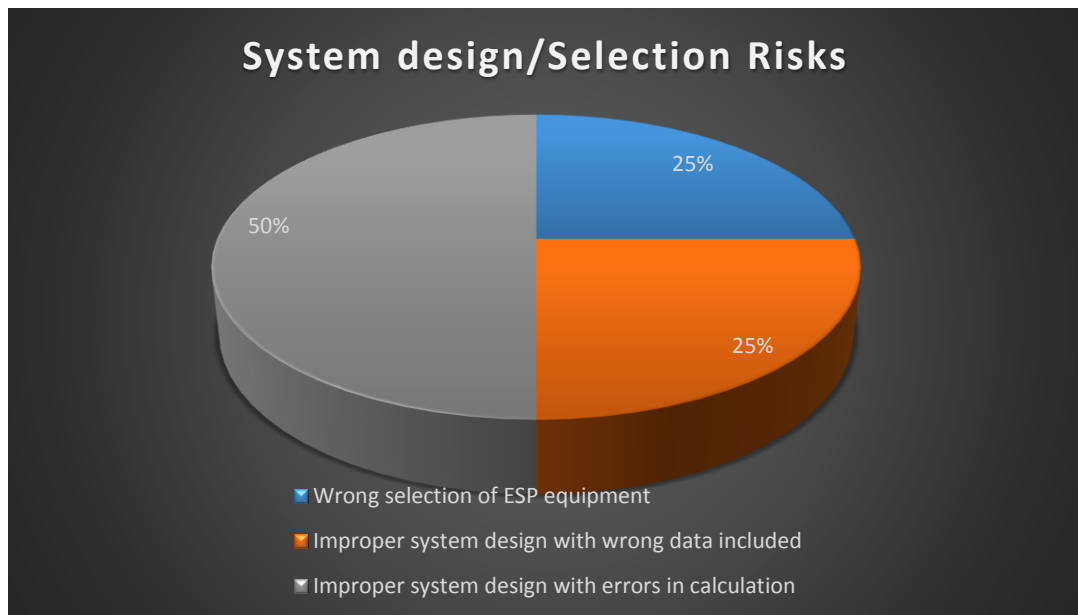


Figure (10)

Installation Risks:

Installation risks are the risks affecting the ESPs objectives and goals.

They include risks arising:

- ✓ Reran damaged equipment.
- ✓ System assembly mounting of the equipment.

By using the 3 x 3 Risk Matrix shown in Table 1 we will assess and evaluate risks based on the probability of occurrence and the impact on the ESPs, we will also assign responsibilities to manage these risks by making residual risks after mitigation actions.

Risk Identification	Risk Analysis	Response Planning	Residual Risk after mitigation actions
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Category	No.of risk	Risk	P	I	Risk Score	Risk Strategy	Responsibility	Risk Mitigation & Actions to resolve	P	I	Risk Score
Installation	1	Reran damaged equipment	2	3	6	Mitigate	ESP technician & Workover CO-Man	Care should made when mounting each cable section together. Proper inspection of the mounted cables. Technician must be equipped with the necessary installation equipment and during installation be placed over the centre of the well.	1	3	3
	2	System assembly Mounting of the equipment	1	2	2	Mitigate	ESP technician & Workover CO-Man	Everyone whose working with the installation is following a thoroughly installation procedure. Rig personnel must be aware that the equipment can easily damage.	1	2	2

Table (5)

Installation risks distribution:

Figure (8) below shows the distribution of the installation risks based on the ratings generated from Table (installation Risk Assessment).

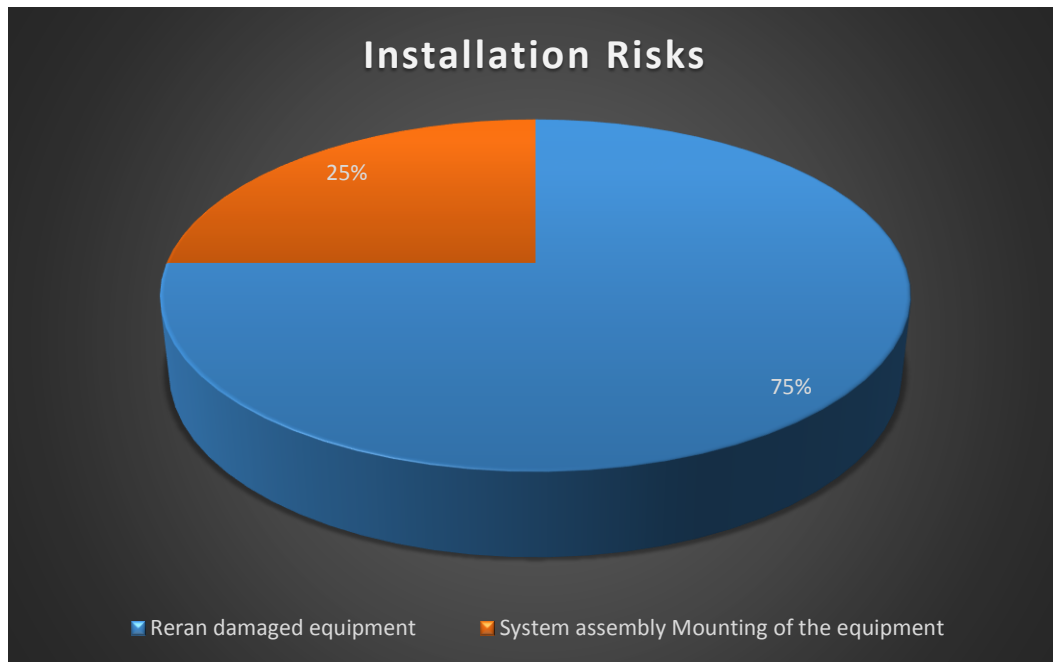


Figure (11)

Installation Plan Risks Responses and Resulting:

Installation risks responses defined as the set of actions that seek to reduce the likelihood of occurrence or the impact of risks, we will examine the different of actions and controls that can employed.

Proper installation of all components is crucial to avoid failures, such as reran damaged equipment and system assembly mounting of the equipment these failures must considered. To reduce these failures it is important that everyone whose working with the installation is following a thoroughly installation procedure. They should have good understanding of handling procedures and adequate

supervision must provide to ensure that the installation operation goes according to plan. Since cable splicing is a major failure cause, care should be made when mounting each cable section together. In addition, proper inspection of the mounted cables is important to prevent faulty installation.

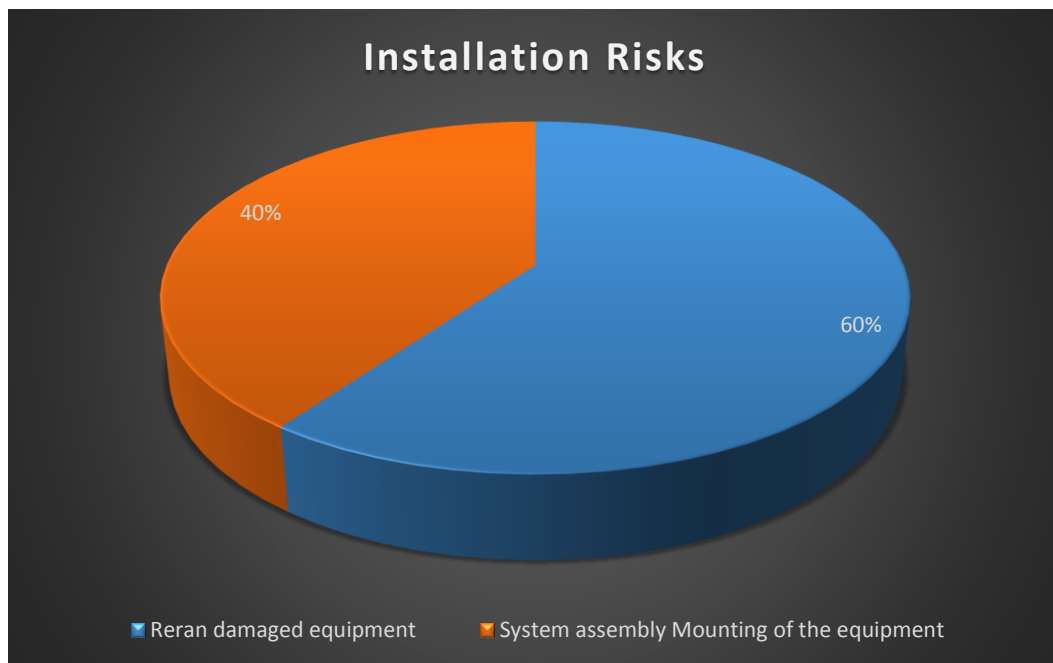


Figure (12)

Manufacturing Risks:

Manufacturing risks are the risks affecting the ESPs objectives and goals.

They include risks arising:

- ✓ Wrong material selection.
- ✓ Improper equipment testing or quality control.

By using the 3 x 3 Risk Matrix shown in Table 1 we will assess and evaluate risks based on the probability of occurrence and the impact on the ESPs, we will also assign responsibilities to manage these risks by making residual risks after mitigation actions.

Risk Identification			Risk Analysis			Response Planning			Residual Risk after mitigation actions		
Category	No.of risk	Risk	P	I	Risk Score	Risk Strategy	Responsibility	Risk Mitigation & Actions to resolve	P	I	Risk Score
Manufacturing	1	Wrong material selection	1	2	2	Mitigate	ESP vendor	Selecting materials that could withstand the corrosive environment. ESP components have been tested for proper operation before it is installed in the well.	1	2	2
	2	Improper equipment testing or quality control	1	3	3	Mitigate	ESP vendor	Full string test, performed both at the manufacturer and on site for quality control.	1	2	2

Table (6)

Manufacturing risks distribution:

Figure (9) below shows the distribution of the installation risks based on the ratings generated from Table (installation Risk Assessment).

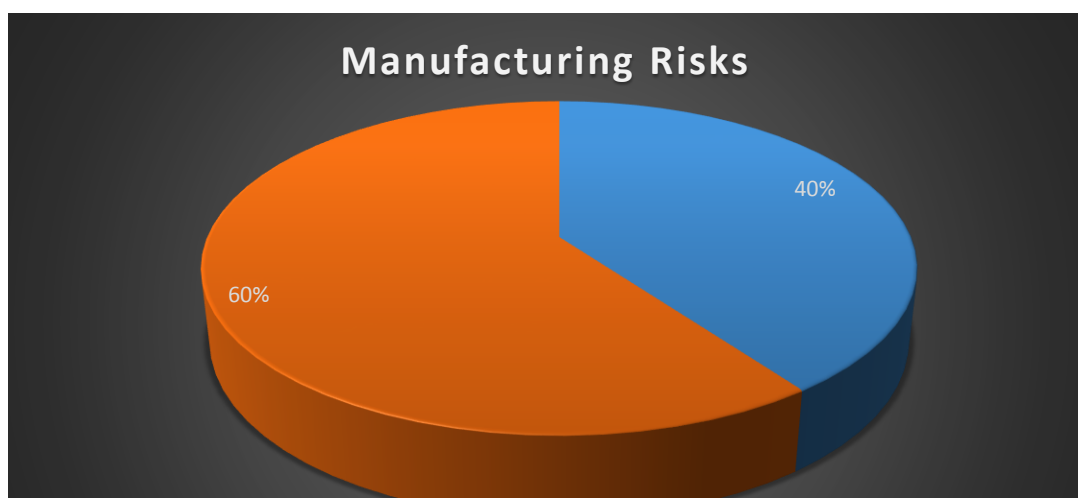


Figure (13)

Manufacturing Plan Risks Responses and Resulting:

Installation risks responses defined as the set of actions that seek to reduce the likelihood of occurrence or the impact of risks, we will examine the different of actions and controls that can employed.

Almost every failure has to do with sand production and corrosion. This give reason think that the materials used have not the right properties to handle the harsh environments.

Wrong material selection: to reduce this problem, Select materials that could withstand the corrosive environment. ESP components have tested for proper operation before install in the well.

Improper equipment testing or quality control: Must testing full string, performed both at the manufacturer and on site for quality control.

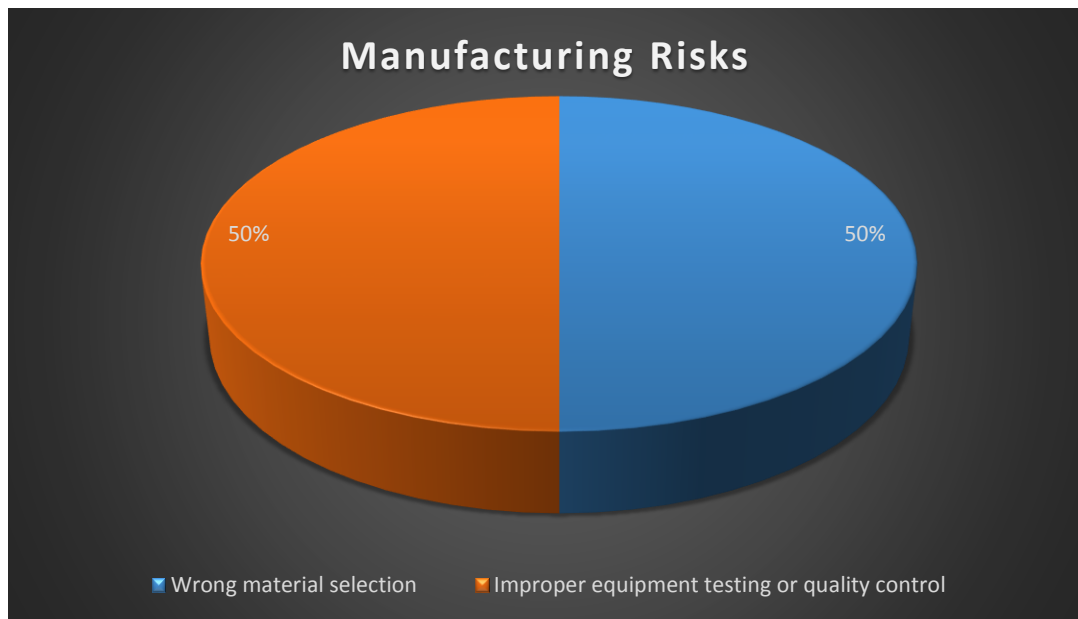


Figure (14)

Conclusion:

This study represents the application of risk management process in Oil Field at Akakus Oil Operations Company. It has been identifying that ESP failures often caused by electric errors because of insufficient protection of the electric motor, pump and cable, which results in a motor failure. The A-field in The Akakus Oil Company will subjected to initial sand production, in addition high production rates and high water cut lead to increased sand production during the field life. Compression pumps handle solids production better than floater pump because it is no contact between impellers and diffusers under operation. Moreover, material selection is an important issue to consider. To improve reliability one has to choose materials, which designated to its application area. This study depending on reducing the cost probability by a factor of two is achievable at a manageable cost to mitigate the ESP failures and reducing Workover operations that leads to high cost.

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